

PATRICK ADRIANUS

Melbourne, Australia | patrickadrianus.com | [LinkedIn](#) | [GitHub](#)
0421806625 | patrickadrianus04@gmail.com

SUMMARY

Data Science graduate (RMIT, 2025) with hands-on experience across the generative AI lifecycle, from dataset curation and LLM benchmarking to on-device model integration and NLP pipeline design. Built and evaluated multi-model prompt configurations using GPT-4, DeepSeek, and Mistral for privacy-issue classification across 10,000+ Stack Overflow posts and shipped a production iOS app integrating on-device small language models (Qwen 3, Llama 3.2) via MLX with deterministic tool routing. Comfortable working across the full ML stack, from raw data preprocessing to model evaluation and deployment. Eager to contribute to generative AI systems that power real-world, user-facing products at scale.

WORK EXPERIENCE

December 2022 – February 2023

Software Development Internship

Urban Waste (Victorian Chamber of Commerce) | Melbourne, Australia

Main Responsibilities:

- Customised and extended Odoo ERP modules using Python, XML, and custom fields to meet specific business requirements across finance, accounting, and logistics functions.
- Designed and documented a data pipeline and database schema diagram to support operational workflows and data consistency.
- Collaborated directly with the founder and cross-functional teams including finance, accounting, and logistics to gather requirements, incorporate feedback, and deliver targeted system enhancements.
- Participated in Agile sprints and translated technical work into clear outcomes for non-technical stakeholders.

PROJECTS

March 2025 – June 2025

LLM – Based Privacy Issues Detection

University Capstone Project

Main Responsibilities:

- Collected 10,000+ Stack Overflow posts via API; manually labelled 1,000+ samples to create a supervised training corpus for privacy-violation detection.
- Benchmarked 5+ LLM model-prompt configurations (GPT-4, DeepSeek, Mistral); DeepSeek Few-Shot achieved best recall (~48%), precision 0.61, F1 0.54, outperforming all rule-based baselines.
- Designed a scalable multi-stage pipeline: keyword filtering → manual labelling → LLM classification, reducing manual review effort significantly.
- Coordinated 4 two-week Agile sprints (JIRA) with 100% on-time delivery; authored client-facing reports and maintained regular supervisor communication.

February 2026 – Present

Blipt – Context-Aware iOS Reminder App

Personal Project

Main Responsibilities:

- Designed and shipped a full SwiftUI + SwiftData iOS app (110+ Swift files, 26 test files) with a coordinator-based scheduling architecture and trace ID logging for reliability and debuggability.
- Built a hybrid NLP parsing engine combining regex, Apple's Natural Language framework, and a Create ML intent classifier trained on 2,640 rows across 12 labels to convert free-text input into structured tasks.
- Integrated on-device small language models (Qwen 3, Llama 3.2) via MLX with deterministic tool routing, retrieving calendar, weather, and location data before prompt injection to improve reliability on smaller models.
- Engineered a Smart Delivery Engine scoring 13 context signals (location, motion, weather, calendar, Wi-Fi, routines) to determine optimal reminder delivery timing.

June 2025 – February 2026

Personal Portfolio Website with AI Assistant

Personal Project

Main Responsibilities:

- Built a full-stack interactive portfolio site using React 19, TypeScript, Vite, and Three.js, featuring GPU-accelerated 3D animations and time-based dynamic theming.
- Integrated a GPT-powered AI chat assistant with portfolio-aware context, capable of answering questions about projects and experience, navigating site sections, and retaining conversation history across sessions.
- Designed a modular service layer separating LLM API logic from UI, with typewriter animation, markdown rendering, and keyboard shortcut support for a polished user experience.

July 2025

CliniWatch – Mental Health App Prototype

RMIT First Health iOS Hackathon (In collaboration with Apple, Northern Health, and Blue)

Goal:

Mental health challenges are a growing global concern, yet deterioration often goes unnoticed between clinical appointments. CliniWatch was built to support early intervention, helping patients build consistent self-monitoring habits to prevent conditions from worsening, while giving carers visibility into progress over time through a dedicated dashboard.

Main Responsibilities:

- Designed and built a local-first iOS mental health companion app in SwiftUI featuring daily mood check-ins, a 7-day mood calendar, gamification system, and a persistent crisis access button with one-tap emergency dialling.
- Implemented a Carer Mode as a first-class UI path, toggling the entire app flow to a dedicated carer dashboard for monitoring patient engagement and accessing support contacts quickly.
- Integrated MapKit and Core Location to surface nearby psychologists, clinics, and mental health services in real time, with full location permission handling.
- Delivered 10+ screens and 15+ SwiftUI views in under 48 hours with a 5-person cross-disciplinary team; earned a Special Mention for Innovation and User Impact from Blue & Northern Health.

July 2024

Pacman Reinforcement Learning Agent

RMIT Academic Project

Main Responsibilities:

- Implemented a tabular Q-Learning agent from scratch using ϵ -greedy exploration and the Bellman update rule, achieving stable navigation and ghost-avoidance policies after sufficient training episodes.
- Extended to Approximate Q-Learning using linear function approximation over hand-engineered features, enabling the agent to generalise across unseen states and scale to larger environments.
- Conducted hyperparameter analysis across learning rate, discount factor, and exploration probability, identifying trade-offs between convergence speed, stability, and long-term reward.
- Validated agent behaviour through win rates and gameplay observation, demonstrating that approximate agents consistently outperformed tabular agents in generalisation.

September 2023

Responsible Generative AI & Model Fairness

RMIT Academic Project

Main Responsibilities:

- Synthesised Australia's e-safety Commissioner Generative AI Position Paper into an executive summary applied to a newsroom deployment scenario, outlining benefits, risks, and governance requirements across the full AI lifecycle.
- Evaluated a trained language model using cross-validation and learning-curve analysis; conducted a formal fairness review covering parity, equalised odds, and inequality index, surfacing data gaps and sources of bias.
- Assessed security, privacy, and ethical considerations end-to-end, from data sourcing and model training through to deployment monitoring and user feedback and delivered a peer review with practical improvements for an ML case in finance.

OTHER EXPERIENCES

March 2023 – Present	Fresh Convenience & Store Services Assistant Main Responsibilities: - Managed store operations including cash handling, inventory, and compliance reporting in a high-volume retail environment. - Led team coordination during manager absences, maintaining service standards and resolving issues independently.
----------------------	--

EDUCATION

September 2025 – Present	IT Professional Year Program ECA
July 2022 – June 2025	Bachelor of Data Science RMIT University Key Study Areas: - Design and evaluate machine learning and deep learning models across classification, reinforcement learning, and time series forecasting tasks. - Build NLP pipelines and assess generative AI systems, including prompt engineering, LLM benchmarking, and responsible AI governance. - Develop scalable data systems spanning big data processing, database design, and software engineering fundamentals. - Apply multivariate analysis, data wrangling, and visualisation to communicate insights to technical and non-technical audiences. Achievements: - Graduated with Distinction (GPA 3.5/4.0) - RMIT University Certificate of Commendation (2023) — Achieved GPA ≥ 3.8 in Bachelor of Data Science - RMIT University Certificate of Recognition (2022) — Awarded for outstanding academic performance (GPA 3.8) in STEM College - Participant, Apple Foundation Program (2024) - Participant, RMIT Hackathon (2025) - Special Mention for Innovation and User Impact — RMIT First Health iOS Hackathon (Bilue & Northern Health, 2025) - Treasurer (Executives), PPIA RMIT (2023–2024) - Event Organiser, PPIA RMIT (2022–2023)

SKILLS SUMMARY

Technical Skills	Soft Skills
Large language models (GPT-4, DeepSeek, Mistral) — benchmarking, prompt engineering, evaluation	Teamwork & adaptability
Machine learning & deep learning (Scikit-learn, PyTorch, TensorFlow)	Cross-functional collaboration
NLP pipeline design (spaCy, Hugging Face, NaturalLanguage framework)	Technical communication
On-device AI inference (MLX, Core ML, Create ML)	Problem-solving
iOS development (Swift, SwiftUI, SwiftData, EventKit, CoreLocation)	Agile delivery & sprint planning
Data analytics & visualisation (Pandas, NumPy, Matplotlib, Seaborn, PowerBI)	Stakeholder reporting
Big data processing (Hadoop MapReduce, Apache Pig)	Attention to detail
Database design — SQL & NoSQL (Oracle, MongoDB)	Time management
Cloud platforms (AWS S3, Lambda)	Critical thinking
Programming — Python, R, Swift, JavaScript, TypeScript, SQL	Interpersonal Skills
Version control & delivery (Git, JIRA, Jupyter, Xcode)	